Unit: Unit 3: Functions **Name:** ______________

Topic: Definition of Relation and Function, Domain, Range, and Notations **Date:** __________

Advanced (Carolina Reaper)

Q1. A bookstore sells x number of books per day. The relation between the number of books sold and the day of the week is given by the table: | Day | Books Sold | | — | — | | Mon | 10 | | Tue | 15 | | Wed | 10 | | Thu | 20 | | Fri | 15 | Is this relation a function?

(A) Yes
(B) No
(C) Maybe
(D) Depends on the day
(E) Not enough information

Q2. The cost C of renting a bike for x hours is given by the equation $C = 5x + 2$. What is the domain of this function?

(A) $xgeq0$
(B) $x > 0$
(C) $xleq0$
(D) $x < 0$
(E) All real numbers

Q3. A water tank can hold 1000 liters of water. The amount of water W in the tank after x minutes is given by the equation $W = 10x$. What is the range of this function?

(A) $0leqWleq1000$
(B) $0 < W < 1000$
(C) $0leqW < 1000$
(D) $0 < Wleq1000$
(E) $Wgeq0$

Q4. A student's grade G is determined by the number of hours x they study per week. The relation between G and x is given by the graph: (the graph shows a straight line passing through (0,0) and (1,10)). Is this relation a function?

(A) Yes
(B) No
(C) Maybe
(D) Depends on the grade
(E) Not enough information

Q5. The distance d between two points (x_1, y_1) and (x_2, y_2) is given by the equation $d = sqrt(x_2 - x_1)^2 + (y_2 - y_1)^2$. What is the range of this function?

(A) $dgeq0$
(B) $d > 0$
(C) $dleq0$
(D) $d < 0$
(E) All real numbers

Q6. A company's profit P is determined by the number of products x they sell. The relation between P and x is given by the equation $P = 2x - 5$. What is the domain of this function?

(A) $xgeq0$
(B) $x > 0$
(C) $xleq0$
(D) $x < 0$
(E) All real numbers

Q7. The cost C of shipping a package is given by the equation $C = 10 + 2x$, where x is the weight of the package in kg. What is the value of C when $x = 5$?

(A) 20
(B) 25
(C) 30
(D) 35
(E) 40

Q8. The height h of a plant after x days is given by the equation $h = 2x + 1$. What is the value of h when $x = 3$?

(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Open Ended Questions (FRQ)

Q9. The price P of a stock is given by the equation $P = 10sin(x) + 20cos(x)$, where x is the time in hours. Find the domain and range of this function and evaluate P when $x =$

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Q10. A water tank can hold 2000 liters of water. The amount of water W in the tank after x minutes is given by the equation $W = 5x+2x^2$. Find the domain and range of this function and evaluate W when $x = 10$. Assume the tank is empty at $x = 0$.

Chef's Tips

- Tip 1: Ensure students understand the concept of relations and functions.
- Tip 2: Emphasize the importance of proper notation.
- Tip 3: Remind students to check their work carefully.